

Relations between fatigue strength and other mechanical properties of metallic materials

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ABSTRACT The relations between fatigue strength and other mechanical properties especially the tensile strength of metallic materials are reviewed. After analyzing the numerous fatigue data available, the qualitative or quantitative relations between fatigue strength and hardness, strength (tensile strength and yield strength) and toughness (static toughness and impact toughness) are established. Among these relations, the general relation between fatigue strength σ_w and tensile strength σ_b , $\sigma_w = \sigma_b(C - P \cdot \sigma_b)$, where C and P are parameters, (hereafter, the general fatigue formula) can well predict the fatigue strength with increasing the tensile strength in a wide range for many materials such as conventional metallic materials, newly developed materials and engineering components. On the basis of the experimental results of many materials, the fatigue damage mechanism, especially for high-strength steels, is proposed. It is suggested that the general fatigue formula can provide a new clue to predict the fatigue strength and design the materials by adjusting material parameters P and C adequately.

Keywords fatigue damage; fatigue strength; fracture mechanism; general fatigue formula; mechanical properties; metallic materials.

NOMENCLATURE

A_K	= impact energy
C, β, m, A, B, P, C and M	= material constants
H	= hardness
HB	= Brinell hardness
HRC	= Rockwell hardness
HV	= Vickers hardness
H_c	= critical hardness
$K_{I_{max}}$	= the maximum stress intensity factor for mode I
N_f	= number of fatigue cycles to fracture
R	= the fatigue ratio of fatigue strength to tensile strength
R_{HRC}	= the ratio of fatigue strength to Rockwell hardness
R_{HV}	= the ratio of fatigue strength to Vickers hardness
R_N	= the ratio of fatigue strength to notch tensile strength
ΔK_{th}	= threshold stress intensity factor range
ε_b	= the maximum tensile elastic strain
σ_a	= stress amplitude
σ_b	= tensile strength
σ_{bc}	= critical tensile strength
σ_{bm}	= notch tensile strength
σ_w^I	= ideal fatigue strength
$\sigma_w^{\max}$	= the maximum fatigue strength
σ_s	= yield strength
σ_w	= fatigue strength
ψ	= contraction of area
$\sqrt{area}$	= the square root of inclusion area

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INTRODUCTION

The emergence of fatigue failure likely dates back to Stone Age during which humans consciously used tools; however, fatigue research had gradually attracted attention until the first half of nineteenth century, when many machine devices had a wide application in mining and railway transport. During the period 1852–1869, Wöhler^{1,2} conducted systematic investigation of steel railway axels, firstly described fatigue behavior by stress amplitude versus cycles to failure (S–N curve), established concept of fatigue limit and found the relationship between fatigue limit and static properties, which laid the foundation for fatigue research and design of metallic materials. In 1910, Basquin^{1,3} found that the S–N curves show a relationship in the finite life region on a log-log plot, which is the famous Basquin relationship. This established empirical relation of S–N curve. With the development of fracture mechanics, research on fatigue of materials could conduct quantitative treatment. In 1980s, Murakami⁴ found that there is a strong correlation between maximum stress intensity factor K_{Imax} and square root of crack area $\sqrt{area}$ obtained by projecting a small defect or crack onto a plane perpendicular to the maximum principal stress, and successively established the simple relation among fatigue strength σ_w , hardness HV and inclusion size $\sqrt{area}$ in high-strength steels. The Murakami relation materially contributed to predicting fatigue strength of high-strength steels.

As described previously, many engineers and scientists have carried out systematic fatigue research for nearly 180 years and attempted to study fatigue mechanism and find simply empirical relationships to predict fatigue properties. For high-cycle fatigue (HCF) and even very high-cycle fatigue (VHCF), fatigue strength or fatigue limit is one of the most important mechanical properties. In S–N curve, some low-strength ferrous alloys can show a horizontal asymptote, that is, fatigue limit, below which there appears to be infinite cycles that will never cause failure; other metallic materials such as high-strength ferrous alloys and non-ferrous metals have no distinct horizontal asymptote and will eventually fail with increasing the number of cycles, for a given number of cycles (e.g. 10^7 or 10^9); the maximum stress amplitude without failure is the fatigue strength. Obviously, fatigue limit can be regarded as a special fatigue strength. Therefore, fatigue strength was used to describe fatigue property in this paper. It is well known that the examination of fatigue strength, especially by staircase method, is very time and money consuming. To find a quick and cost-effective way, many attempts have been made to determine the fatigue strength.^{2,5–9} Some empirical relations relating fatigue strength to other mechanical properties, such as yield strength (YS),¹⁰ tensile strength^{7,11,12} and hardness^{4,13,14} can be found elsewhere. However, it is

still lacking systematic investigation on the relations between fatigue strength and other mechanical properties. Therefore, this gives rise to some questions for scientists and engineers: what mechanical properties do have relations with fatigue strength? Among those empirical equations, is there a more universal equation to predict fatigue strength of metallic materials? If so, what is the fatigue mechanism to control the general equation? Therefore, in this paper, we carried out some systematic mechanical experiments and collected many fatigue data and equations available to summarize the simple relationship between fatigue strength and other mechanical properties of metallic materials, seeking for the general fatigue relation and corresponding fatigue mechanism.

RELATION BETWEEN FATIGUE STRENGTH AND HARDNESS

Hardness is the most simply tested mechanical property; on the contrary, fatigue strength is probably the most difficultly tested one. It is willing to predict the fatigue strength by hardness. Actually, there are some empirical relations^{5,7,14–16} between fatigue strength σ_w and hardness used widely in engineering as following:

$$\sigma_w \cong 1.6HV \pm 0.1HV \quad (HV \leq 400), \quad (1)$$

$$\sigma_w \approx 1.6HB \quad (HB \leq 400), \quad (2)$$

$$\sigma_w \approx 0.25HB: \quad (HB \leq 400) \quad (3)$$

where σ_w is in megapascal in Eqs. (1) and (2); σ_w is in kip per square inch in Eq. (3); HV and HB are kilogram-force per square millimetre in Eqs. (1–3). In addition, the factors of proportional relation between σ_w and HB are 1.72 and 1.43, reported from Refs [17] and [18], respectively.

Equations (1–3) are applicable when hardness (HV or HB) is less than 400; but not conservative when hardness (HV or HB) is more than 400.

Garwood *et al.*⁵ investigated fatigue properties of several high-strength alloy steels and also found that the fatigue strength increases linearly at first and then decreases with hardness further increasing, as shown in Fig. 1. Therefore, it is difficult to determine the relation between fatigue strength and hardness for steels in the high-strength level, which is owing to the influence of inclusion size and location in steels.^{4,14} To solve the question, Murakami⁴ masterly designed, systematically performed many fatigue experiments and then proposed a simple and useful equation to predict the fatigue strength by Vickers hardness and the size and location of inclusion as follows:

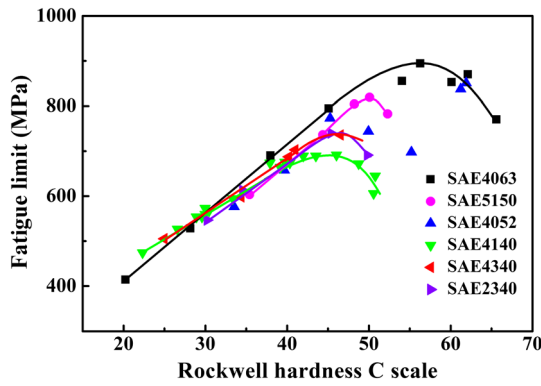


Fig. 1 Relation between Rockwell hardness and fatigue limits of several alloy steels.⁵

$$\sigma_w = \frac{C(HV + 120)}{(\sqrt{area})^{1/6}}. \quad (4)$$

$\sqrt{area}$ is in micrometre; C is the material constant depending on crack initiation site and for surface, subsurface and inner inclusion equals to 1.43, 1.41 and 1.56, respectively. Equation (4) can be used to predict effectively fatigue strength of high-strength steels by hardness and inclusion size and location obtained from experiment. Therefore, Eq. (4) has a wide application to study the fatigue properties of high-strength steels. Hereafter, some modified Murakami formulae were proposed.

Wang *et al.*¹⁹ found Eq. (4) has a relation with a specified number of cycles to failure N_f and can be written as

$$\sigma_w = \frac{\beta(HV + 120)}{(\sqrt{area})^{1/6}} \quad (5)$$

where the constant $\beta = 3.09\text{--}0.12 \log N_f$ for interior inclusions, and $2.79\text{--}0.18 \log N_f$ for surface inclusions or defects.

Based on the effect of hydrogen during fatigue crack initiation, Liu *et al.*²⁰ developed an equation as follows:

$$\sigma_w = 2.7 \frac{(HV + 120)^{15/16}}{(\sqrt{area_{in}})^{3/16}}. \quad (6)$$

McGreevy *et al.*²¹ combined the Hall–Petch relation and Murakami model and proposed HPM (Hall–Petch–Murakami) model. Xu *et al.*²² found that Eq. (4) also can apply to extruded magnesium alloy. We²³ found that Eq. (4) cannot accurately estimate the fatigue strength of copper–beryllium (Cu–Be) alloy failed from inclusion. At present, it is not reported that Eq. (4) can apply other metals except high-strength steels and the magnesium alloy. Moreover, there is no equation to

predict the fatigue strength of metallic materials with a wide range of hardness value. To find the equation, we investigated the hardness and fatigue properties of AISI 4340 steel and Cu–Be alloy with a wide strength range and summarized their relations between hardness and fatigue strength as the following:

Experimental results of AISI 4340 show that VHCF strength firstly increases and then decreases with increasing hardness, which is in accordance with the results of other steels performed by Garwood *et al.* mentioned previously. After further analysis, it is interesting to find that with increasing hardness, the fatigue ratio R_{HV} or R_{HRC} gradually declines in a linear relation; obviously, the fatigue strength σ_w can be written in a quadratic equation of the hardness HV or HRC , as shown in Fig. 2. The fitting equations can be expressed as follows:

$$R_{HV} = 2.42 - 2.21 \times 10^{-3} HV \quad (\text{AISI 4340 steel}), \quad (7)$$

$$\sigma_w = (2.42 - 2.21 \times 10^{-3} HV) \cdot HV \quad (\text{AISI 4340 steel}), \quad (8)$$

$$R_{HRC} = 23.84 - 0.22 HRC \quad (\text{AISI 4340 steel}), \quad (9)$$

$$\sigma_w = (23.84 - 0.22 HRC) \cdot HRC \quad (\text{AISI 4340 steel}). \quad (10)$$

It can be seen that the fatigue ratios and fatigue strengths are basically within the 5% error band. This illustrates that Eqs. (8) and (10) can well describe the relation between fatigue strength and hardness of the AISI 4340 steel in a wide hardness range.

The experimental results of Cu–Be alloy strengthened by precipitation indicate that the fatigue strength also increases at first and then decreases with the increase of hardness; it is amusing that the relation between fatigue strength and hardness is quadratic (Fig. 3a). The equation can be written as follows:

$$\sigma_w = (2.41 - 5.34 \times 10^{-3} HV) \cdot HV \quad (\text{Cu – Be alloy}). \quad (11)$$

It can be seen that the fatigue strength predicted by Eq. (11) in Fig. 3a are basically within the 10% error band. This indicates that the quadratic equation can decently describe the relation between fatigue strength and hardness of the two samples, implying that the

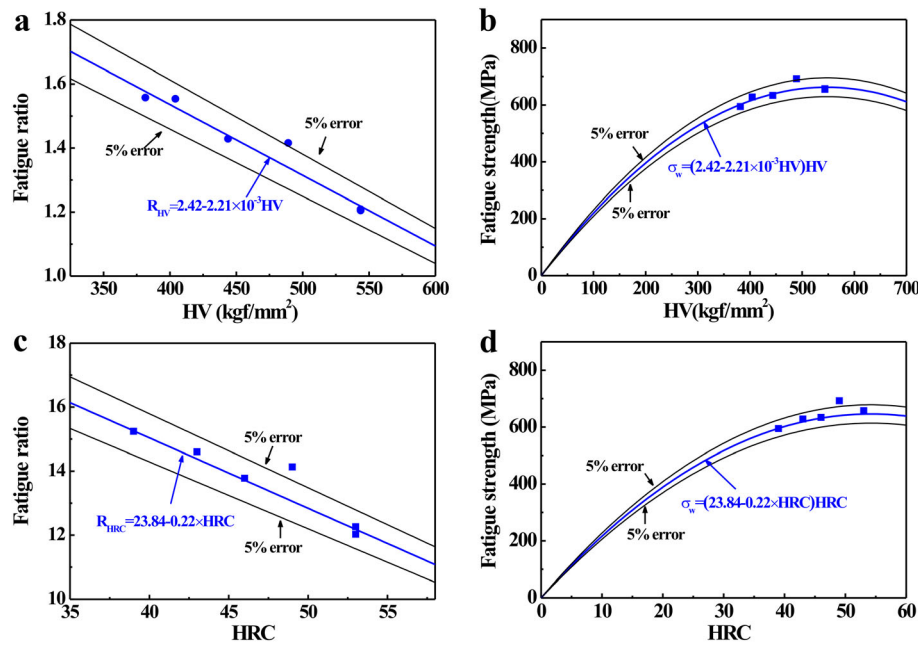


Fig. 2 Relations between hardness and very high-cycle fatigue properties of the AISI 4340 steel: (a) Vickers hardness (HV) and fatigue ratio, (b) Vickers hardness and fatigue strength, (c) Rockwell hardness (HRC) and fatigue ratio, and (d) Rockwell hardness and fatigue strength.

quadratic fatigue equation also suits for metallic materials processed by the same strengthening technology.

It is curious for us that whether the quadratic relation suits other metallic materials. Thus, we collected some fatigue data of several high-strength alloy steels, such as SAE 4340, 4140, 2340, 4063⁵ and SUP7.²⁴ Figs 3b–f show the relation between rotating bending fatigue strength and hardness of some high-strength steels. Similar with the relation of AISI 4340 and Cu–Be alloy, the quadratic relations between fatigue strength and HRC (or HV), respectively are the following:

$$\sigma_w = (24.07 - 0.16HRC) \cdot HRC \quad (\text{SAE4063}), \quad (12)$$

$$\sigma_w = (24.73 - 0.19HRC) \cdot HRC \quad (\text{SAE4340}), \quad (13)$$

$$\sigma_w = (20.43 - 0.09HRC) \cdot HRC \quad (\text{SAE5150}), \quad (14)$$

$$\sigma_w = (27.39 - 0.28HRC) \cdot HRC \quad (\text{SAE4140}), \quad (15)$$

$$\sigma_w = (2.80 - 2.35 \times 10^{-3}HV) \cdot HV \quad (\text{SUP7}). \quad (16)$$

It is seen from Fig. 3 that error bands of fatigue strength nearly are less than 5%; this indicates that quadratic equation can apply well to the five high-strength steels.

After summarizing the fatigue data of seven specific materials (AISI 4340, Cu–Be alloy, SUP7, SAE 4340,

4140, 2340, and 4063), a conclusion can be drawn: the relation between fatigue strength and hardness is quadratic. Furthermore, Eqs. (8) and (10)–(16) can be unified in a more universal formula as follows:

$$\sigma_w = (A - B \cdot H) \cdot H \quad (17)$$

where H is hardness; A and B are two constants that for some metallic materials are listed in Table 1. Moreover, the values of the maximum fatigue strength $\sigma_w^{\max}$ and critical hardness H_c where the maximum fatigue strength reached can be derived from Eq. (17) as follows:

$$\sigma_w^{\max} = \frac{A^2}{4B}, \quad (18)$$

$$H_c = \frac{A}{2B}. \quad (19)$$

It is obvious that $\sigma_w^{\max}$ and H_c are dependent on constants A , and B . According to their values, the $\sigma_w^{\max}$ and corresponding H_c can be estimated, as listed in Table 1. Their error bands of the fatigue strength almost are within the 5%, which indicate that the Eq. (17) can accurately predict the fatigue strengths of metallic materials treated by the same kind of processing technology.

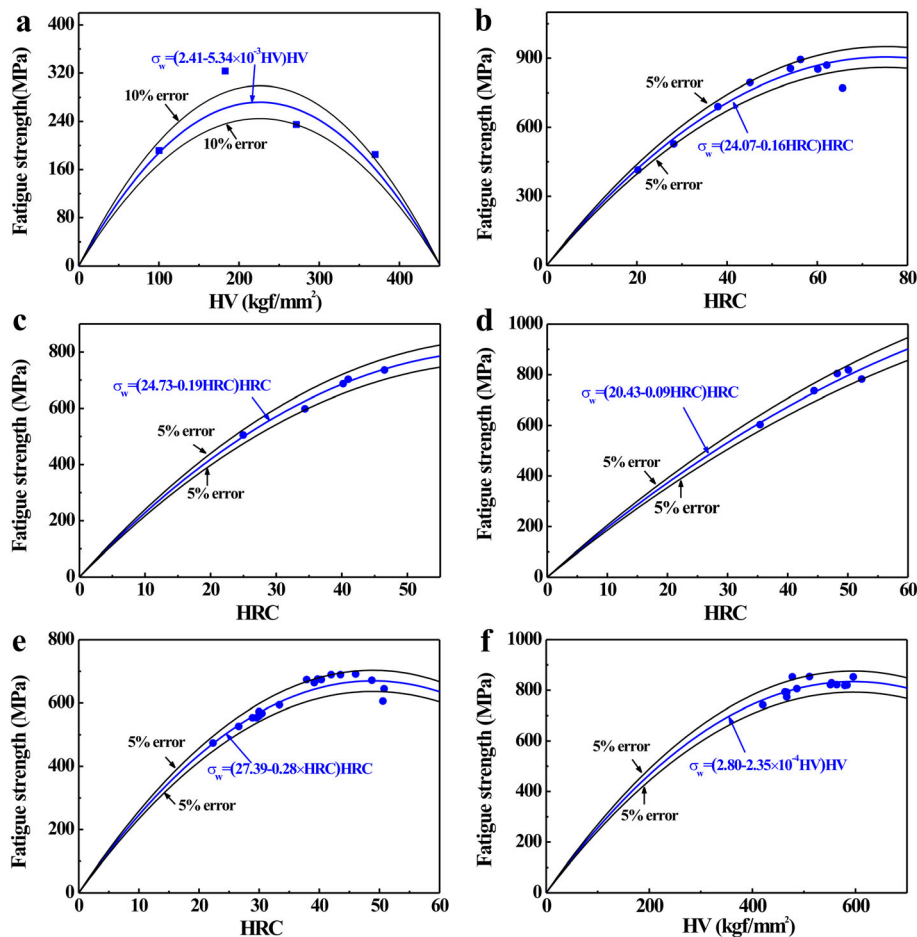


Fig. 3 The fitting relations between hardness and fatigue strength of metallic materials: (a) Cu–Be alloy, (b) SAE 4063 steel, (c) SAE 4340 steel, (d) SAE 5150 steel, (e) SAE 4140 steel and (f) SUP7 steel. (Rotating bending fatigue data of SAE 4063, 4340, 5150 and 4140 from Ref. [5]; rotating bending fatigue data of SUP7 steel from Ref. [25]) HV, Vickers hardness; HRC, Rockwell hardness.

Table 1 Values of the constants A , B and corresponding H_c and $\sigma_w^{\max}$

Hardness	Material	A	$B/10^{-3}$	H_c	$\sigma_w^{\max}/\text{MPa}$	Error band (%)
HV	AISI 4340	2.42	2.21	548	662	± 5
	QBe2	2.41	5.34	226	272	± 5
	SUP7	2.80	2.35	596	834	± 5
HRC	AISI 4340	23.84	0.22	54	646	± 5
	SAE 4063	24.07	0.16	75	905	± 5
	SAE 4340	24.73	0.19	65	805	± 5
	SAE 5150	20.43	0.09	114	1159	± 5
	SAE 4140	27.39	0.28	49	670	± 5

H_c : critical hardness, above which fatigue strength does not increase. For Vickers hardness (HV), B : mm^2/kg ; H_c : kg/mm^2 ; and for Rockwell hardness (HRC), B and H_c : no unit.

RELATION BETWEEN FATIGUE STRENGTH AND TENSILE STRENGTH

As mentioned previously, hardness has certain relations with fatigue strength; we wish to confirm whether there

is a quantitative relation between tensile strength and fatigue strength.

In 1860s, Wöhler firstly found that the relation between fatigue strength σ_w and tensile strength σ_b is simply linear,²

$$\sigma_w = (0.4 \sim 0.5) \sigma_b. \quad (20)$$

It also is the first quantitative relation between fatigue strength and other mechanical properties.

After 1950s, many high-strength engineering metallic materials such as steels, copper and aluminum alloys had been well developed. After analyzing the numerous fatigue data of those materials,^{7,11,12} we can sum up a more general form in lower strength level,

$$\sigma_w = m \sigma_b. \quad (21)$$

However, in higher strength level, the fatigue strength keeps constant or decreases with further increasing the tensile strength.^{11,12} This is to say, the linear relation Eq. (21) does not apply to materials in high-strength level. However, relevant Murakami equations mentioned previously can be used for high-strength steels. In a word, there is also no suitable formula to satisfactorily express the general relation between tensile and fatigue strengths of materials with a wide strength range. To solve the question, we investigated AISI 4340 steel⁹ and Cu-Be alloy^{23,25} with a very wide tensile strength range.

Experimental results of AISI 4340 steel heat-treated under different conditions show the VHCF strengths increase first and then decrease with increasing tensile strengths. After fitting by software, it was found that the fatigue ratio R , which is accordant with R_H mentioned previously, gradually declines as tensile strength increases in a linear relation,

$$R = 0.70 - 1.85 \times 10^{-4} \sigma_b. \quad (22)$$

As shown in Fig. 4a, the error band is within 5%. Obviously, the fatigue strength σ_w can be written in a quadratic equation as function of the tensile strength σ_b as follows:

$$\sigma_w = (0.70 - 1.85 \times 10^{-4} \sigma_b) \cdot \sigma_b. \quad (23)$$

Equation (23) is drawn in Fig. 4b; it can be seen that the fatigue strength is basically within the 5% error band. This indicates that Eq. (23) can well estimate the relation between fatigue strength and tensile strength of the AISI 4340 steel in a wide strength range.

The experimental results of Cu-Be alloys^{23,25} indicate that the relation between HCF strength and tensile strength is quadratic (Fig. 5a). The equation can be written, respectively, as follows:

$$\sigma_w = (0.54 - 3.11 \times 10^{-4} \sigma_b) \cdot \sigma_b. \quad (24)$$

Their fatigue strength almost appears within 5% error band. This illustrates that the quadratic equation also is suitable for Cu-Be alloy after different ageing technologies. For confirming the quadratic fatigue strength equation above, we searched many fatigue data of metallic materials available.

At first, the rotating bending fatigue data of some high-strength alloy steels with large-scale strength levels were collected from Ref. [5]. For SAE 4140, 4340, 2340 and 4063 steels, the fitting relations between fatigue strength and tensile strength shown in Figs 5b-e are also quadratic. The fitting equations for fatigue strength are as follows, respectively.

$$\sigma_w = (0.87 - 2.65 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{SAE 4140}), \quad (25)$$

$$\sigma_w = (0.76 - 1.78 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{SAE 4340}), \quad (26)$$

$$\sigma_w = (0.74 - 1.89 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{SAE 2340}), \quad (27)$$

$$\sigma_w = (0.92 - 2.37 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{SAE 4063}). \quad (28)$$

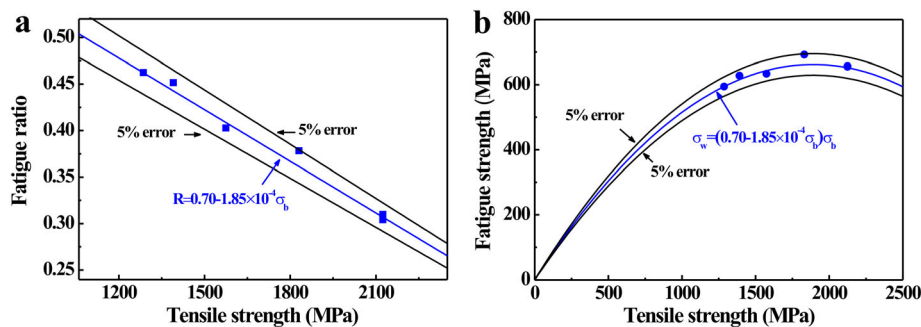


Fig. 4 Relations between tensile strength and very high-cycle fatigue properties of the AISI 4340 steel: (a) tensile strength and fatigue ratio, and (b) tensile strength and fatigue strength.

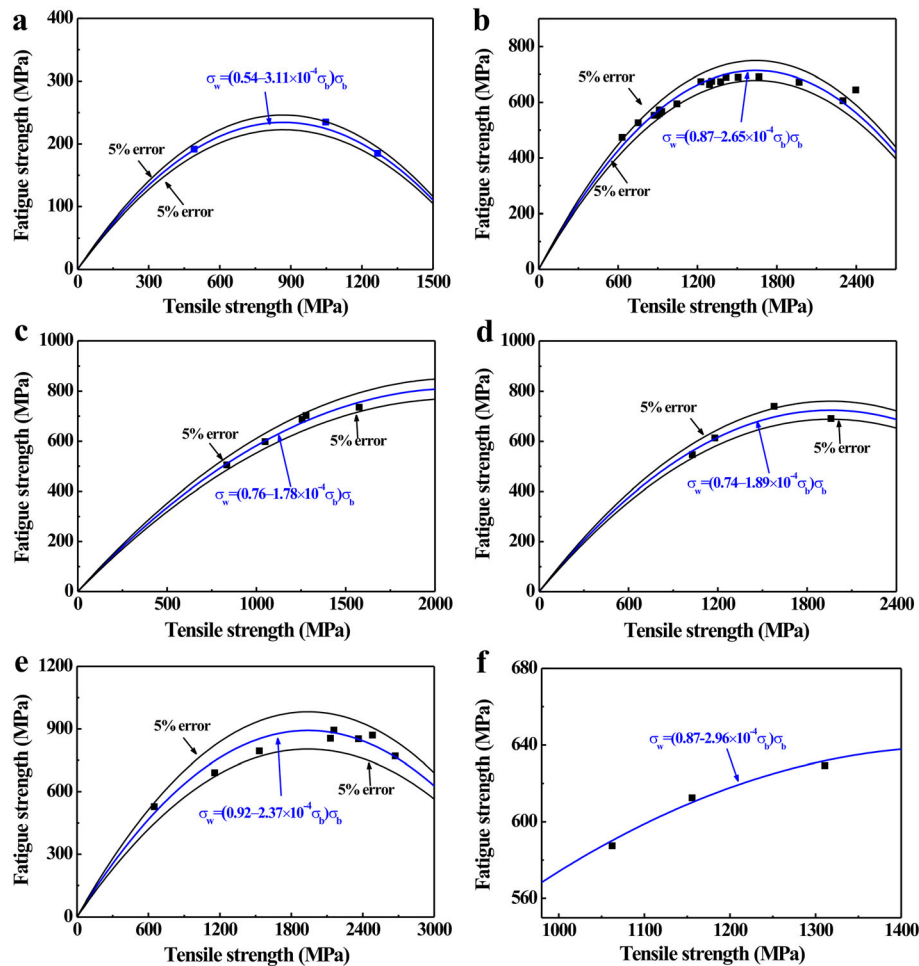


Fig. 5 The fitting relations between tensile strength and fatigue strength of metallic materials: (a) Cu–Be alloy,²⁵ (b) SAE 4140 steel, (c) SAE 4340 steel, (d) SAE 2340 steel, (e) SAE 4063 steel and (f) 1023Ti alloy. (Data of Cu–Be alloy are collected from Ref. [25]; data of SAE 4140, 4340, 2340 and 4063 are cited from Ref. [5]; and data of Ti alloy are collected from Ref. [26].)

Equations (25)–(28) are drawn in Fig. 5, it is obvious that the error band is basically within 5%. This suggests that the quadratic equations can be used to predict fatigue strengths by tensile strengths of those steels.

To further verify whether the relation between fatigue strength and tensile strength is quadratic, we collected the data of a titanium alloy 1023 from Ref. [26]. It is obvious from Fig. 5f that this alloy meets the quadratic equation,

$$\sigma_w = (0.87 - 2.96 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{Ti 1023}). \quad (29)$$

Equation (29) can accurately predict the relation between fatigue strength and tensile strength, because the error band is less than 5%. Therefore, Fig. 5 shows that the fatigue strength formulae Eqs. (24)–(29) previously are also applicable to other metallic materials with a very wide range of tensile strength.⁹

Secondly, we collected many more fatigue data of engineering materials, such as alloy steels, aluminum alloys, copper alloys and titanium alloys.¹¹ Fig. 6 exhibits the fitting curves of their fatigue strengths, and the fitting equations can be expressed as the following, respectively,

$$\sigma_w = (0.59 - 9.24 \times 10^{-5} \sigma_b) \cdot \sigma_b \quad (\text{For alloy steels}), \quad (30)$$

$$\sigma_w = (0.54 - 3.72 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{For copper alloys}), \quad (31)$$

$$\sigma_w = (0.53 - 5.66 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{For aluminum alloys}), \quad (32)$$

$$\sigma_w = (0.62 - 1.09 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{For titanium alloys}). \quad (33)$$

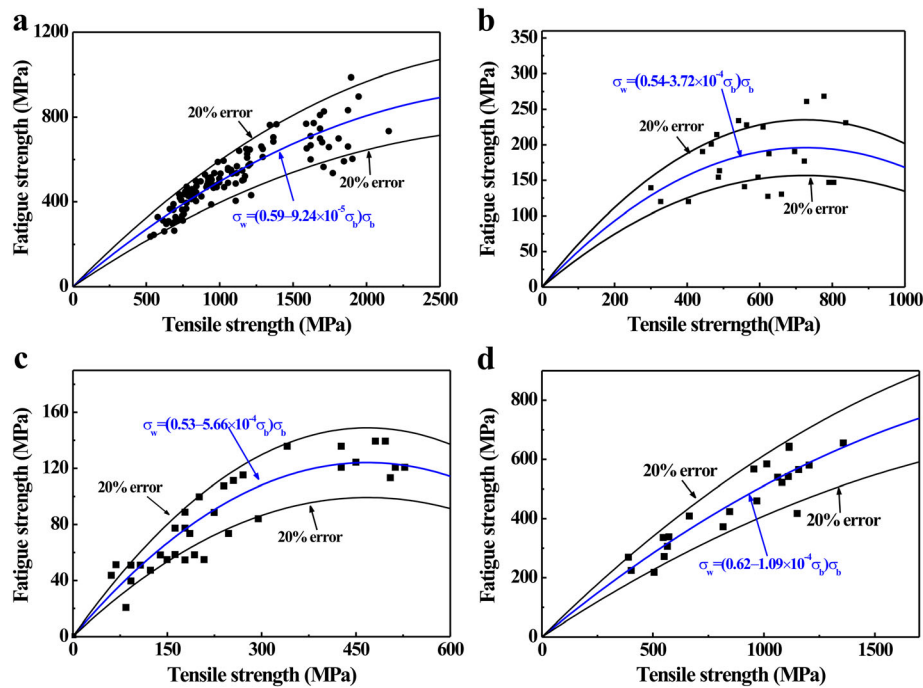


Fig. 6 The fitting relations between tensile strength and rotating bending fatigue strength: (a) for wrought steels at 10^7 to 10^8 cycles, (b) for copper alloys at 10^8 cycles, (c) for aluminum alloys at 5×10^8 cycles, (d) for titanium alloys at 2×10^7 cycles. (Data are collected from Ref. [11].)

In Fig. 6, fatigue strength data mostly are within the 20% error band. Besides, Ref. [9] reported that many alloying steels (such as 60Si2Mn, 42CrMo, 42CrMoVNb and 40CrNiMo) under very high frequency (VHF) fatigue condition and the wrought steels also satisfy the quadratic equation; the equations can be written as follows:

$$\sigma_w = (0.67 - 1.52 \times 10^{-4} \sigma_b) \sigma_b \quad (\text{For VHF fatigue}), \quad (34)$$

$$\sigma_w = (0.61 - 1.24 \times 10^{-4} \sigma_b) \sigma_b \quad (\text{For wrought steels}). \quad (35)$$

This indicates that the quadratic formulae Eqs. (30–35) previously are also appropriate for the conventional engineering materials.

Thirdly, many new high-strength materials such as nanostructured and ultrafine-grained (UFG) materials have been successfully fabricated in the last 30 years. We curiously wondered whether the quadratic relation can apply to them or not. However, there are a few fatigue data of the identical materials with a very wide strength range. Fortunately, we have performed some fatigue tests on UFG low-carbon steel,²⁷ and collected some fatigue data of coarse-grained^{28,29} and UFG^{29,30} ones and fitted them as shown in Fig. 7a. The relations between tensile strength and fatigue strength can be written as follows:

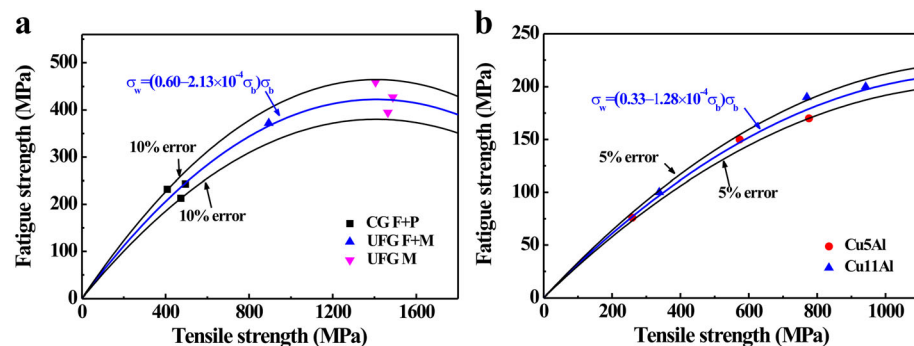


Fig. 7 The fitting relations between tensile strength and fatigue strength of new developed metallic materials: (a) coarse grained (CG) and ultrafine-grained (UFG) low-carbon steels, and (b) UFG Cu–Al alloys (Data of steels are collected from Refs. [28]–[30]; and data of Cu–Al alloys are collected from Ref. [31].)

$$\sigma_w = (0.60 - 2.13 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{For low carbon steel}). \quad (36)$$

The data of fatigue strength are also within the 10% error band. In addition, we also have collected and analysed the data of UFG copper–aluminum alloys³¹ and found that the relation between fatigue strength and tensile strength is quadratic (Fig. 7b) as follows:

$$\sigma_w = (0.33 - 1.28 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{For Cu – Al alloy}). \quad (37)$$

In Fig. 7b, it is known that fatigue strengths of Cu-5Al and Cu-11Al mainly are inside the 5% error band, which indicates that the quadratic formula previously is also applicable to the ultrafine and nanostructured metallic materials.

Fourthly, notice that the general fatigue formula previously is proposed and verified on the basis of smooth specimens under the stress ratio of $R = -1$. Actually, there are some factors affecting the prediction of fatigue strength, including surface condition, specimen size, loading mode and so on. Small changes of those factors can also obviously change the fatigue properties, making analytical prediction of fatigue life or strength difficult. Therefore, it is necessary to confirm whether the general fatigue formula can apply to engineering components.

After collecting many references, we only found a few related fatigue data. Fig. 8a shows the relation between tensile strength and fatigue strength of a hand-operated gear-shift lever with various types of surface.³² Fatigue data of the component under ground, machined, hot-rolled and forged states were fitted by the quadratic fatigue formula as follows:

$$\sigma_w = (0.48 - 5.13 \times 10^{-5} \sigma_b) \cdot \sigma_b \quad (\text{Ground}), \quad (38)$$

$$\sigma_w = (0.44 - 9.46 \times 10^{-5} \sigma_b) \cdot \sigma_b \quad (\text{Machined}), \quad (39)$$

$$\sigma_w = (0.37 - 1.44 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{Hot – rolled}), \quad (40)$$

$$\sigma_w = (0.28 - 1.22 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{Forged}). \quad (41)$$

By comparison, their relations between tensile strength and fatigue strength suit well the quadratic fatigue formula.

Fig. 8b presents the influence of shot peening on the fatigue strength of high-strength steel¹⁶ and a shaft³ made of another steel. Their fatigue data before and after shot peening were fitted by the quadratic fatigue formula. The equations are the following:

$$\sigma_w = (0.75 - 2.95 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{Steel, before}), \quad (42)$$

$$\sigma_w = (0.72 - 1.09 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{Steel, after}), \quad (43)$$

$$\sigma_w = (0.67 - 2.62 \times 10^{-4} \sigma_b) \cdot \sigma_b \quad (\text{Shaft, before}), \quad (44)$$

$$\sigma_w = (0.68 - 6.04 \times 10^{-5} \sigma_b) \cdot \sigma_b \quad (\text{Shaft, after}). \quad (45)$$

It is seen from Fig. 8b that the quadratic fatigue formula can accurately predict the relation between tensile strength and fatigue strength. The experimental results in Fig. 8 verify the general fatigue formula can apply the engineering components.

In summary, we analysed the fatigue data of eight specific materials (AISI4340, Cu–Be alloy, SAE 4340, 4140, 2340, 4063, Ti alloy 1023, Cu–Al alloy), four types

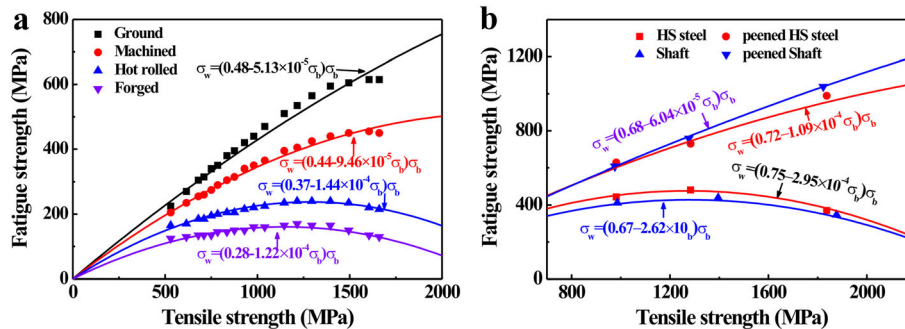


Fig. 8 Relations between tensile strength and fatigue strength of components and materials treated by different technologies: (a) a hand-operated gear-shift lever with various type of surface and (b) a high-strength (HS) steel and a shaft processed by shot peening. (Fatigue data of the hand-operated gear-shift lever were from Ref. [32]; fatigue data of high strength were collected from Ref. [16]; and fatigue data of a shaft were collected from Ref. [4].)

of materials (steels, copper alloys, aluminum alloys and titanium alloys) previously and some engineering components (materials) treated by different technologies and drew the general conclusion: the relation between fatigue strength and tensile strength is quadratic. Therefore, Eqs. (23–45) can be expressed in the general form as follows:

$$\sigma_w = (C - P \cdot \sigma_b) \cdot \sigma_b \quad (46)$$

where C and P are two constants; moreover, each material has its own constants as listed in Tables 2 and 3. If introducing Young's modulus E of materials into Eq. (46), one can obtain

$$\sigma_w = (C - M \cdot \sigma_b / E) \cdot \sigma_b = (C - M \cdot \varepsilon_b) \cdot \sigma_b \quad (47) \quad \sigma_w^{\max} = \frac{C^2}{4P} = \frac{EC^2}{4M}, \quad (48)$$

where $M = P \cdot E$; ε_b is schematically illustrated in Fig. 9a. $\sigma_b \cdot \varepsilon_b$ in Eq. (47) represents the twice elastic strain energy per unit volume before necking. Therefore, it is known that the simple schematic illustration of the relation between tensile strength and fatigue factors can be shown in Fig. 9b: $\varepsilon_b \cdot \sigma_b$ represents the rectangle a–b–c–d; $M \cdot \varepsilon_b \cdot \sigma_b$ represents the rectangle a–b–f–e; $C \cdot \sigma_b$ represents the rectangle a–b–g–h; therefore, the area of the rectangle e–f–g–h can be considered as the fatigue strength.

From Eqs. ((46)) and (47), the values of the maximum fatigue strength $\sigma_w^{\max}$ and critical tensile strength σ_{bc} over which fatigue strength begins to decrease can be derived as follows:

Table 2 Values of the constants C , P , M and E and corresponding σ_{bc} and $\sigma_w^{\max}$ of metallic materials

Material	C	$P/10^{-4}$ MPa $^{-1}$	M	E GPa	σ_{bc} MPa	$\sigma_w^{\max}$ MPa	Error (%)
AISI 4340	0.70	1.85	38.5	208	1891	662	±5
SAE 4140	0.87	2.65	54.9	207	1642	714	±5
SAE 4340	0.76	1.78	37.0	208	2135	811	±5
SAE 2340	0.74	1.89	39.1	207	1958	724	±5
SAE 4063	0.92	2.37	49.1	207	1941	893	±5
Steel ^a	0.67	1.52	31.5	207	2204	738	±15
Steel ^b	0.61	1.24	25.7	207	2460	750	±20
Steel ^c	0.59	0.92	19.1	207	3207	946	±20
LC steel ^c	0.60	2.13	45.3	208	1408	423	±10
QBe2	0.54	3.11	40.4	130	868	234	±5
Cu alloy	0.54	3.72	44.3	119	726	196	±20
Cu–Al alloy	0.33	1.28	13.1	102	1289	213	±5
Al alloy	0.53	5.66	40.6	71.7	468	124	±20
1023 Ti	0.87	2.96	32.6	110	1470	639	<±5
Ti alloy	0.62	1.09	12.0	110	2844	882	±20

Steel

^aVery high frequency fatigue under tension and compression; steel. ^bRotating bending fatigue; steel. ^cRotating bending fatigue from another reference; LC steel: low-carbon steel.

Table 3 Values of the constants C , P , M , σ_{bc} and $\sigma_w^{\max}$ of materials and components treated by different technologies

Material	Treating technology	C	$P/10^{-4}$ MPa $^{-1}$	σ_{bc} MPa	$\sigma_w^{\max}$ MPa
Gear-shift lever	Grinding	0.48	0.513	4678	1123
	Machining	0.44	0.946	2326	512
	Hot rolling	0.37	1.44	1285	238
	Forging	0.28	1.22	1148	161
High-strength steel		0.75	2.95	1271	477
	Shot peening	0.72	1.09	3303	1189
shaft		0.67	2.62	1279	428
	Shot peening	0.68	0.604	5629	1914

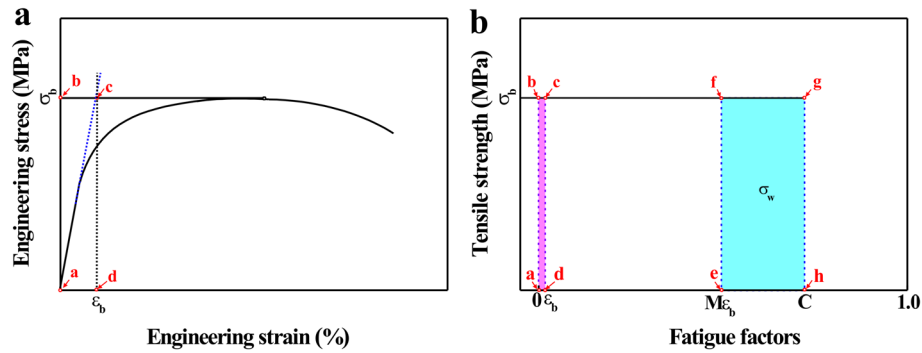


Fig. 9 Schematic representation of relation between fatigue strength and tensile engineering stress–strain curves: (a) illustration of tensile elastic strain and (b) illustration of fatigue strength according to the general fatigue formula.

$$\sigma_{bc} = \frac{C}{2P} = \frac{EC}{2M}. \quad (49)$$

From Eqs. (48) and (49), it can be concluded that $\sigma_w^{\max}$ and σ_{bc} are dependent on C , P or E , and C and M . According to their values, the corresponding $\sigma_w^{\max}$ and σ_{bc} can be estimated, as listed in Tables 2 and 3. Compared with the error bands in Table 2, the error bands of one type of materials such as steels, copper, aluminum and titanium alloys are within 20%, which are the maximum. The error bands of many steels under the same experimental condition are within 15%; and within 10% for low-carbon steel with very wide span of grain size, within 5% for a specific material (SAE 4140, 4340, 2340 and 4063, Cu–Be, Cu–Al, Ti 1023), which is the minimum. This indicates that the less are the influencing factors of material's composition and production, the more accurate prediction of the formula.

For the quadratic equation, the fatigue strength and the maximum fatigue strength can be designed or improved by adjusting the constants P (M) and C . As to the mechanism and method, we will discuss in detail at the later section.

RELATION BETWEEN FATIGUE STRENGTH AND SOME OTHER MECHANICAL PROPERTIES EXCEPT HARDNESS AND TENSILE STRENGTH

There are many data and references in which the relations between fatigue strength and hardness or tensile strength are involved as mentioned previously. Are there still some general relations between fatigue strength and other properties? For answering the question, we collected and listed some relations as follows.

Notch tensile strength

We investigated the mechanical properties of AISI 4340 steel and found that as tensile strength increases, fatigue ratio R_N linearly increases and fatigue strength quadratically changes (Fig. 10); their equations are the following, respectively:

$$R_N = 0.48 - 8.7 \times 10^{-5} \sigma_{bm} \quad (\text{Notch tension}), \quad (50)$$

$$\sigma_w = (0.48 - 8.7 \times 10^{-5} \sigma_{bm}) \cdot \sigma_{bm} \quad (\text{Notch tension}). \quad (51)$$

Their error bands almost are within 5%. The tendency is accordant with that of smooth sample as shown

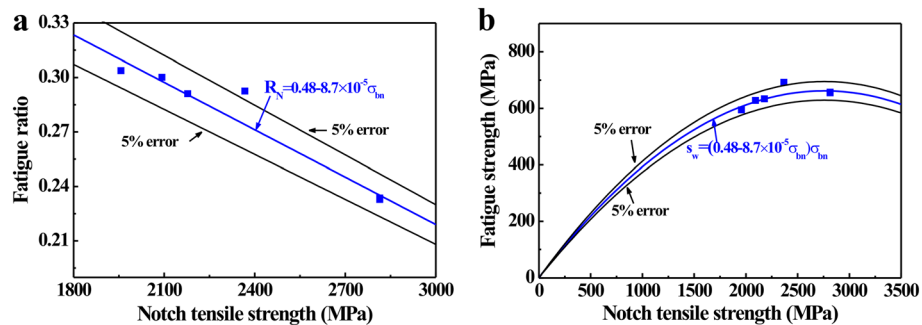


Fig. 10 Relations between notch tensile strength and very high-cycle fatigue properties of the AISI 4340 steel: (a) notch tensile strength and fatigue ratio, and (b) notch tensile strength and fatigue strength.

in Fig. 4. We regretfully found no other materials to verify the relation. At present, we do not know whether the quadratic relation between fatigue strength and notch tensile strength is an individual or common case.

Yield strength

In addition, we also found that the relation between VHCF strength and YS σ_y in AISI 4340 steel is linear (Fig. 11a),

$$\sigma_w = 0.22\sigma_y + 335. \quad (52)$$

Fatigue data almost are within 5% error band, which indicate that Eq. (52) can describe well the relation between fatigue strength and YS of AISI 4340. However, the linear relation does not fit for copper beryllium alloy, as displayed in Fig. 11b; there are two fatigue data beyond 20% error band. Moreover, some Refs [6,15] reported that there is no obviously quantitative relation between fatigue strength and YS; for some engineering materials,^{7,11,12} UFG materials^{30,33,34} and discontinuously reinforced aluminum alloy,³⁵ fatigue strength does not relate to YS, but tensile strength in high-cycle or VHCF regime. In theory, slip in grain can lead to fatigue crack initiation, and YS has a relationship with the start of slip in grains; therefore, YS is thought to have a strong relation with fatigue strength. In fact, the fatigue strength for high cycle or very high cycle is the threshold stress for crack propagation and not the critical stress for crack initiation. Therefore, the linear relation between fatigue strength and YS of AISI 4340 is probably a special case. The reason of the special case is that, for ultra-high-strength steels, with tensile strength increasing, YS increases firstly and then, decreases with increasing tensile strength (Fig. 12), which is a general case reported in some Refs [9,36]. Therefore, the relation between YS and tensile strength is very complex, and then, the relation

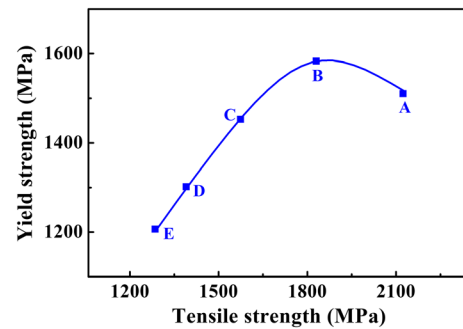


Fig. 12 Relation between yield strength and tensile strength of AISI 4340.⁹

between YS and fatigue strength is too complex to find a general relation.

Impact toughness

We analysed the relation between fatigue strength and impact toughness of AISI 4340 steel and copper beryllium alloy and found that there is no obvious relation, as shown in Figs. 13a–c. Some impact toughness data of AISI 4340 are very scattered in the lower toughness region indicated by an ellipse as shown in Fig. 13a. The relations between fatigue ratio and impact toughness as well as between fatigue strength and impact toughness of copper beryllium alloy are not linear and quadratic, respectively as shown in Figs 13b and c, which only told us that the high or low impact toughness cannot lead to the highest fatigue strength. However, Lin *et al.*³⁷ systematically investigated the relation between rotary bending fatigue strength and impact energy of a number of different grades of austempered ductile irons (ADIs) and found the relations are linear, as shown in Fig. 13d.

$$\sigma_w = 116.50 - 1.35 \cdot A_k \quad (\text{ADI} - A), \quad (53)$$

$$\sigma_w = 180.86 - 0.84 \cdot A_k \quad (\text{ADI} - C), \quad (54)$$

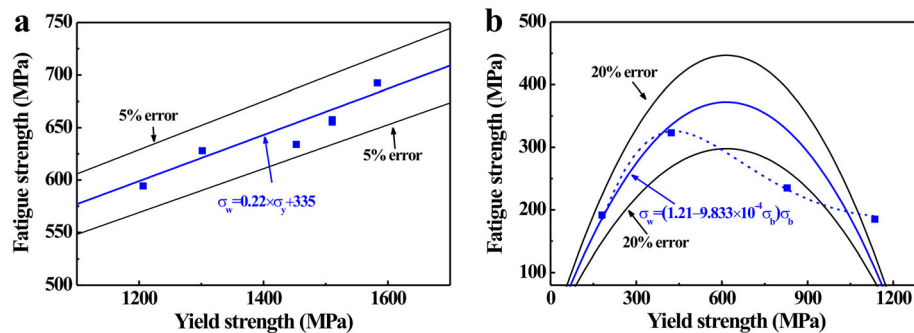


Fig. 11 The fitting relations between yield strength and fatigue strength: (a) AISI 4340 steel under condition of very high-cycle fatigue and (b) copper–beryllium alloy under condition of high-cycle fatigue.

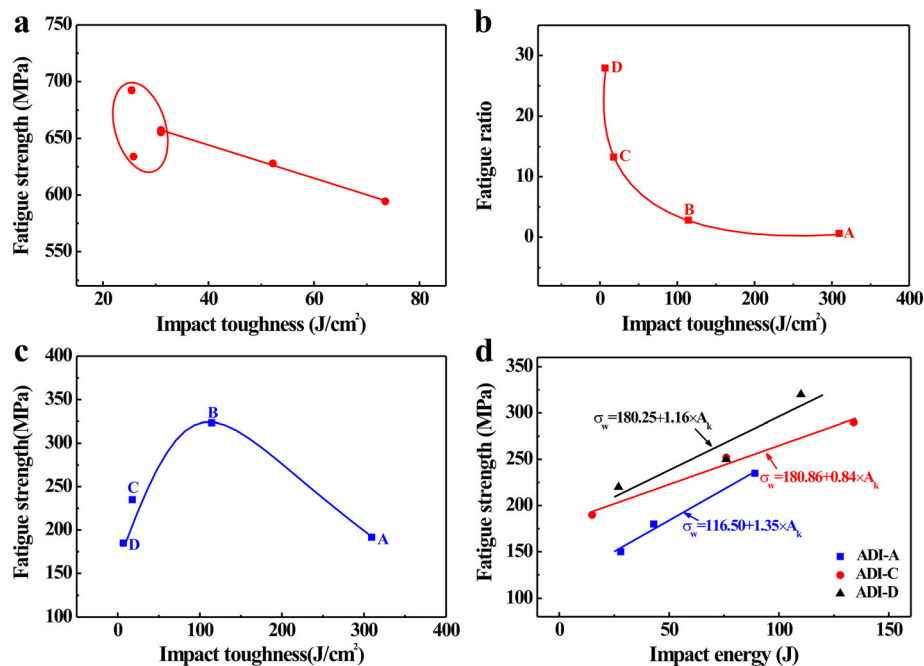


Fig. 13 The fitting relations between impact properties and fatigue strength of metallic materials: (a) AISI 4340 steel under condition of very high-cycle fatigue, (b) and (c) Cu-Be alloy under condition of high-cycle fatigue and (d) austempered ductile irons (ADI) under condition of high-cycle fatigue (data of ADI from Ref. [37]).

$$\sigma_w = 180.25 - 1.16 \cdot A_k \quad (\text{ADI - D}). \quad (55)$$

For ADIs – A, C and D, greater amounts of retained austenite and higher toughness lead to a better HCF strength, which only apply to ADI. It needs more experimental data to verify whether the linear relation can be used to other materials.

Tensile toughness

As we all known, static toughness is an easily measured parameter, which can be simply obtained from tensile stress–strain curve. Ye *et al.*³⁸ established a power-law relation between static toughness and low-cycle fatigue properties. Enlightened by the new approach, we tried to find quantitative relation between fatigue strength and static toughness from AISI 4340 steel and copper beryllium alloy. According to Ref. [39], we defined three types of tensile toughness: TA, the product of ultimate tensile strength (UTS) and elongation to fracture; TB, the product of YS and uniform elongation; and TC, the product of elastic limit and uniform elongation. The parameter of TA approximately stands for the strain energy per unit volume prior to fracture where severe plastic deformation including voids formation occurs; TB represents approximately the appreciable strain energy per unit volume before necking where large plastic deformation including huge amount of dislocation collapse occurs; and TC represents roughly the

maximum elastic strain energy per unit volume before necking where less dislocations just operate locally.

It can be seen from Fig. 14a: For Cu-Be alloy, TA decreases with increasing UTS and TB decreases more slowly with increasing UTS; however, TC increases a little and then decreases with increasing UTS. Among all the specimens, the specimen with tensile strength of 670 MPa has the highest toughness (TC), this might imply that the high fatigue strength of Cu-Be alloy is associated with the higher toughness TC. Compared Fig. 14a with the fitting relation between fatigue properties and tensile strength of Cu-Be alloy in Ref. [23], a relatively better correlation between fatigue ratio and toughness TC can be found in current Cu-Be alloy. It is obvious that the higher toughness TC, the higher fatigue ratio is for this copper alloy. The reason may be understood as follows. In the HCF regime (10^7 cycles), the applied cyclic stress is in the elastic stress range. Therefore, the toughness TA and TB cannot correlate the fatigue properties well due to severe or large plastic deformation involved. However, the fatigue crack initiates at defects where stress is just yielding and dislocations operate locally; the toughness TC can correlate this process relatively well.

However, it can be found from Fig. 14b: For AISI 4340 steel, TA gradually increases as UTS increases; TB and TC decrease at first and then, gradually increase with tensile strength increasing. Tendencies of TA, TB and TC with tensile strength are different from those of

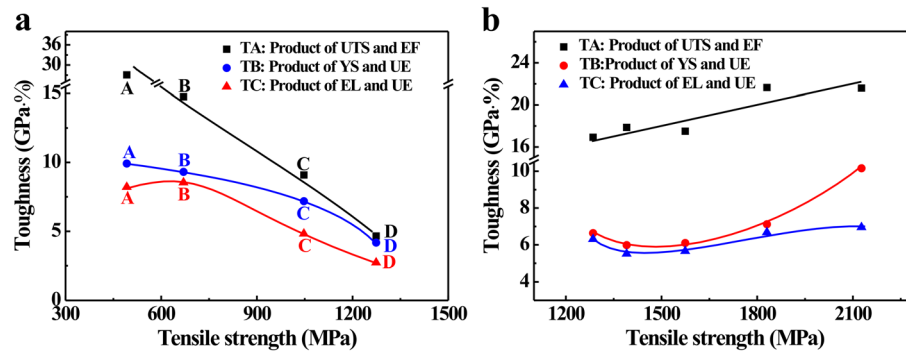


Fig. 14 Relations between different tensile toughnesses (TA: the product of ultimate tensile strength (UTS) and elongation to fracture (EF); TB: the product of yield strength (YS) and uniform elongation (UE), and TC: the product of elastic limit (EL) and UE) and tensile strength: (a) Cu–Be alloy and (b) AISI 4340.

Cu–Be alloy in Fig. 14a. Compared with Figs 4 and 14b, no correlation between fatigue ratio (or fatigue strength) and toughness TC can be found in steel as in current Cu–Be alloy. Now, because few fatigue data about the relation between fatigue properties and tensile toughness are found. We cannot make sure whether the relation is a common or individual case.

Other empirical equations

Besides, there are some fatigue empirical relations^{7,40} as follows:

$$\sigma_w = 0.23(\sigma_y + \sigma_b) \quad (\text{Structural steel}), \quad (56)$$

$$\sigma_w = 0.25(1 + 1.35\psi)\sigma_b \quad (\text{High-strength steel}), \quad (57)$$

$$\sigma_w = 1.49\sigma_b^{0.63} \quad (\text{Aluminum alloy}). \quad (58)$$

By substituting mechanical data of AISI 4340 steel⁴¹ into Eqs. (56) and (57), as shown in Table 4, it is found that error band in Eq. (56) is within 5%; an error band in Eq. (57) is more than 25%, which indicates that by comparison, Eq. (56) is more suitable for the steel. The reason is similar with that of Eq. (52) on the relation

between fatigue strength and YS. The mechanical data of Cu–Be alloy²⁵ show that YS increases continuously with tensile strength increasing; therefore, Eq. (56) is not suitable for the Cu–Be alloy. They were substituted into Eq. (57), the fatigue strengths of Cu–Be alloy are 219, 287, 344 and 347 MPa, and the corresponding error bands are 4.25%, –11.1%, 46.52% and 87.74%. Obviously, Eq. (57) is not applicable to the Cu–Be alloy either from the general trend. This implicates that both Eqs. (56) and (57) are not common relations.

By substituting fatigue data of aluminum alloys in Fig. 6c into Eq. (58), we can obtain that the fatigue strength estimated by Eq. (58) is less than experimental value; even the fatigue data are obviously different from the estimated results in the general trend, which means that Eq. (58) cannot apply the aluminum alloys with a wide range of strength.

After comparing all fatigue equations mentioned previously, we can summarize that the relations between fatigue strength and hardness, specially between fatigue strength and tensile strength are applicable to many types of materials such as engineering materials (steel, copper, aluminum and titanium alloys), newly developed materials (fine-grained low carbon steel and Cu–Al alloy) and engineering components; the fatigue equations related with notch tensile strength, YS, impact toughness,

Table 4 The prediction of fatigue strength of AISI 4340 steel by empirical formula

σ_b , MPa	2127	1830	1574	1390	1285
σ_y , MPa	1510	1583	1452	1301	1206
ψ	0.4	0.51	0.46	0.48	0.51
σ_w , MPa, Exp.	655	693	634	628	594
σ_w , MPa [Eq.(56)]	652	661	643	616	594
σ_w , MPa [Eq.(57)]	837	785	696	619	573
Error [Eq.(56)], %	–0.47	–4.6	1.5	–2.0	0
Error [Eq.(57)], %	28	13	9.8	–1.4	–3.5

Exp.: experimental value; $Error = (\sigma_{w,eq.} - \sigma_{w,exp.}) / \sigma_{w,exp.}$.

tensile toughness and so on only suit well for one or two materials. Because the monotonic tensile loading to determine tensile properties is part of the tension–compression cyclic loading to determine fatigue strength, and it is known that the compressive loading has minor effect on fatigue damage, we prefer to figure out the relation between fatigue strength and tensile strength, and named Eq. (46) as the general fatigue formula. Fatigue mechanism related to the general fatigue formula was discussed as follows.

FATIGUE MECHANISM TO CONTROL THE GENERAL FATIGUE FORMULA

The real materials inevitably have different kinds of defects,^{4,7,42} which profoundly influence the mechanical properties of metals. For example, the strengthening nature of metals can be attributed to the interactions of dislocations with point defects (atomic vacancies and interstitials), strengthening particles (solute atoms, precipitates and dispersed particles), boundaries (grain boundaries, phase boundaries and twin boundaries) and dislocations. Some defects can be intentionally used to improve the mechanical properties of a material. However, some defects are destructive for mechanical properties. For example, casting defects such as porosity, shrinkage and cavity can lead to deterioration in the mechanical behavior of the products.⁴³ In addition, some defects such as inclusion and scratch almost do not influence the static properties, but obviously influence fatigue properties of high-strength metals. In principle, fatigue cracks can nucleate at different sites, such as slip bands, deformation bands, large-angle grain boundaries, shear bands and phase boundaries.^{44,45} Fatigue crack preferentially initiates at the area with the highest stress (stress concentration or local plastic deformation). For soft materials with coarse grains, such as pure copper⁴⁶ and annealed 0.13% carbon steel,⁴ fatigue failure frequently originates from persistent slip bands for HCF, even for VHCF. In UFG materials, fatigue crack originates from shear bands.⁴⁷ Twin boundaries may be important crack nucleation sites in hexagonal close-packed materials⁴² such as magnesium, titanium and zinc.⁴⁸ Inclusions and second-phase particles are commonly the dominant nucleation sites in materials with commercial purity such as aluminum alloys⁴² and high-strength steels.^{12,42,49} In addition, there are some special initiation sites, such as at soft-phase ferrite in a ferrite and martensite due-phase steel,⁵⁰ intergranular facet of high-strength steel⁵¹ and cleavage crack of titanium alloy.⁵² In fact, there may be many different kinds of defects such as grain boundary, twin boundary, phase boundary, inclusion and so on in a real material at same time as mentioned previously;

however, the fatigue cracking is generally originated from one type of them depending on the loading condition and other factors. This implicates that there must be competition of defects, especially when the strength changes, the competition will lead to a visible transition of fatigue cracking sites. For example, fatigue cracking sites of AISI 4340 steel are gradually transformed from surface to the inner inclusions,⁹ the similar tendency of transition from surface to interior cracking is roughly described with tensile strength increasing in the Ref. [53]. When fatigue cracks are initiated from inner inclusion for all samples, fatigue strength begins to decrease with increasing tensile strength. For Cu–Be alloy,²⁷ at relatively low-strength level, some fatigue cracks originated from defects; whereas at high-strength level, all the fatigue cracks initiated from cleavage facets. When all fatigue samples are initiated from cleavage facets, fatigue strength also begins to decrease with increasing tensile strength. In summary, the influences of defects on mechanical properties are very complicated: some influences are beneficial, others are harmful; some influences are negligible on static properties, but are very important on fatigue property; only on fatigue property, the influences of some defects are principal for low-strength metals, but negligible for high-strength metals.

For simplicity, in Ref. [9], defects can be roughly categorized into two types: (1) the intrinsic defects (IDs) such as vacancies, interstitial atoms, dislocations, stacking faults and grain boundaries (GB) are related to the materials itself and hardly to be completely avoided by controlling preparation technology; (2) the processing defects (PDs), including non-metallic inclusions, cavities, segregation and scratches, and so on, are associated with the technology of production or processing and can be avoided to some extents.

Firstly, for the low-strength coarse-grained materials, fatigue cracks almost do not nucleate at PDs (e.g. inclusion or scratch), but at slip bands (SBs) or GBs⁴⁴ on the sample surface (Fig. 15). The materials with IDs and without PDs can be regarded as ideal ones, because ID mainly controls the microstructure and determines the main mechanical properties. Generally speaking, static mechanical properties (tensile strength and YS) almost are not related with PDs, and the fatigue strength σ_w might only depend on its IDs, and the corresponding ideal fatigue strength σ_w^I should show a proportional relation with its tensile strength σ_b as follows:

$$\sigma_w^I = C \cdot \sigma_b. \quad (59)$$

The following example of grain-refinement strengthening was given to explain the relation Eq. (59).

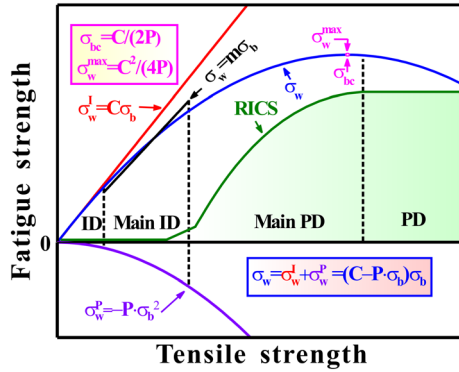


Fig. 15 Schematic illustration of fatigue mechanisms and strength formula with tensile strength increasing.⁹ In the fatigue-strength-tensile-strength plot, the relations between fatigue strength and tensile strength are explained and the transition of the fatigue mechanism from the intrinsic defect (ID) controlled to processing defect (PD) controlled with tensile strength increasing is indicated.

As we known, the smaller grain size is, the larger will be the total boundary surface area per unit volume, which will enhance the pile-up of dislocations, make the motion of dislocation difficult and result in the increase of tensile strength, as explained by the well-known Hall–Petch relation. Therefore, the relation between tensile strength and grain size $D^{-1/2}$ is linear and can be written as

$$\sigma_b \propto \frac{1}{\sqrt{D}}. \quad (60)$$

As for fatigue strength, small-sized grains can lead to the reduced flaw sizes and increase difficulties for the imposed stress concentration at the flaw to exceed the critical stress of the material, thus suppressing early crack nucleation and propagation. Fatigue strength also follows the Hall–Petch relation^{34,54} as follows:

$$\sigma_w \propto \frac{1}{\sqrt{D}}. \quad (61)$$

Therefore, the relation between tensile strength and fatigue strength is linear, which can be confirmed by the experimental results of some UFG materials in Refs [33,34].

Secondly, for the high-strength materials, fatigue cracks seldom initiate along SB or GBs and other IDs, but at PDs (inclusion or scratch in Fig. 15). Now, the fatigue strength of materials is considered as PD-controlled, and the effects of IDs on fatigue strength could be minified to some extent. As first approximation, the defect is assumed as a crack embedded in the material; therefore, the fracture mechanics can be used. By which, Murakami *et al.*⁴ found a

well-known fatigue strength formula as mentioned previously. If the area of a crack at inclusion is denoted by *area*, then the maximum value, $K_{I\max}$ of the stress intensity factor along its crack front is obtained approximately by⁵⁵

$$K_{I\max} = 0.5 \sigma_a \sqrt{\pi \sqrt{area}} \quad (62)$$

Here, $K_{I\max}$ is in megapascal metre^{1/2}, σ_a is in megapascal, and $\sqrt{area}$ is in metre. The threshold for crack growth can be written as⁴

$$\Delta K_{th} = 3.3 \times 10^{-3} (HV + 120) (\sqrt{area})^{1/3} \quad (63)$$

where ΔK_{th} is in megapascal metre^{1/2} and *HV* is in kilogram-force per square millimetre, and $\sqrt{area}$ is in micrometre.

When driving force $K_{I\max}$ equals to resisting force ΔK_{th} , the fatigue strength can be determined. Combining Eqs. (62) and (63), noticing the difference of the unit of the $\sqrt{area}$ between those equations, the fatigue strength σ_w can be expressed as Eq. (4), which is widely used to predict the fatigue strength of high-strength steel with inclusion as mentioned previously. Whereas the fatigue strength measured by staircase method is a statistical value, if the specimens were prepared with the same batch of steel bars but different heat treatments, the average inclusion sizes contained in those specimens should be close to each other. Therefore, the inclusion size $\sqrt{area}$ in Eq. (4) can be considered as a constant; thus, Eq. (4) becomes

$$\sigma_w = C'_0 (HV + 120) \quad (64)$$

where $C'_0 = \frac{C_0}{(\sqrt{area})^{1/6}}$. In practice, the hardness *HV* and tensile strength σ_b roughly show linear relation Eq. (21) as presented previously; so that Eq. (64) is approximately consistent with Eq. (59). It implies that the relation between tensile strength and fatigue strength of one material, no matter whether in low-strength level or in high-strength level, is still linear in theory.

However, a question was raised: For our fatigue experiments, all specimens were cut from the same batch of steel bars but different heat treatments; the inclusion size contained in those specimens should be the same from the statistical viewpoint. Therefore, the influence of the inclusion size on fatigue strength of those specimens could be neglected. According to Eqs. (59) or (64), the fatigue strength should be proportional to the tensile strength at low-strength and high-strength levels; however, it is contradictory with the experiment results indicated by Eq. (46). To explain this, the derivation of

Eq. (4) needs be analysed again. The driving force $K_{I\max}$ is obtained by fracture mechanics, which should be alright; the resisting force ΔK_{th} is based on the empirical relations as the following:

$$\Delta K_{th} \propto (\sqrt{area})^{1/3} \quad (65)$$

and

$$\Delta K_{th} \propto (HV + C). \quad (66)$$

If Eq. (4) is not appropriate, ΔK_{th} expressed by previous equations must be not suitable. Because Murakami *et al.*⁴ mainly consider the influence of inclusion size $\sqrt{area}$ and neglected the influence of the wide strength range. For this, we collected many fatigue data and found the relation between ΔK_{th} and tensile strength σ_b of 300 M steel in Ref. [56], as shown in Fig. 16; ΔK_{th} linearly decreases with increasing tensile strength with a wide range. The linear relation can be written as

$$\Delta K_{th} \propto C_1 - C_2 \cdot \sigma_b \quad (67)$$

where C_1 and C_2 are material constants. The decrease of ΔK_{th} can be also attributed to the enhancement of defect sensitivity with tensile strength increasing. Therefore, ΔK_{th} should be modified. Combining Eqs. (65–67) leads to the new ΔK_{th} ,

$$\Delta K_{th} = C(HV + 120)(\sqrt{area})^{1/3} (C'_1 - C'_2 \sigma_b). \quad (68)$$

Combining Eqs. (62) and (68), the fatigue strength formula can be written as

$$\sigma_w = \frac{C'_0(HV + 120)(C'_1 - C'_2 \sigma_b)}{(\sqrt{area})^{1/6}}. \quad (69)$$

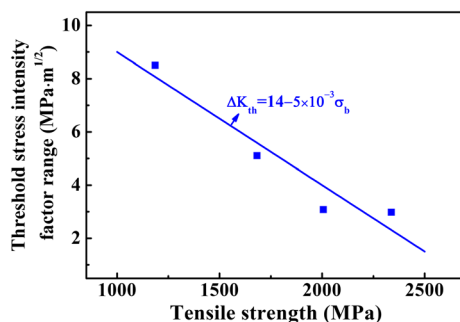


Fig. 16 Relation between threshold stress intensity factor range ΔK_{th} at $R = 0.05$ and tensile strength of 300 M steel tempered by different temperatures, adapted from Ref. [56].

Assuming the inclusion size as a constant in average in all fatigue tests, and the linear relation between tensile strength σ_b and hardness HV as mentioned previously, Eq. (69) can be further simplified as Eq. (46), that is, $\sigma_w = (C - P \cdot \sigma_b) \cdot \sigma_b$. This indicates that to a first approximation, the general fatigue formula can be explained by fracture mechanics.

In this section, we only gave an example of AISI 4340 steel. In a word, the decrease in the fatigue strength at high-strength level can be explained by fracture mechanics and attributed to the transition of fatigue cracking sites from surface to the inner inclusions, resulting in the maximum fatigue strength $\sigma_w^{\max}$ at an adequate tensile strength level, which was explained in detail in Ref. [9]. In spite of many materials suiting for general fatigue formula mentioned previously, we cannot know clearly their microstructure and the types of fatigue cracking sites from existing references. There will be many different fatigue cracking mechanisms. However, we can conclude that with changing tensile strength, the transition of fatigue cracking mode controlled by fatigue mechanism leads to the variation of fatigue strength as the general fatigue formula depicted.

SUMMARY

With the development and research of near 180 years, many quantitative equations and models for fatigue strength have been proposed. To find a simple and accurate fatigue strength equation, we collected many data including fatigue properties. By analysis and comparison, we found that the relations between fatigue strength and hardness, especially between fatigue strength and tensile strength display quadratic equation; the relations between fatigue strength and notch tensile strength, YS, impact toughness and tensile toughness separately do not show a general relation.

The general fatigue strength formula $\sigma_w = \sigma_b(C - P \cdot \sigma_b)$ can well describe the fatigue strength of conventional engineering materials (steels, copper and aluminum, and titanium alloys), newly developed materials (UFG and nanocrystalline materials) and some components (hand-operated gear-shift lever and shaft) with increasing the tensile strength, and predict the maximum fatigue strength and the corresponding tensile strength.

On the basis of many fatigue data, we proposed the fatigue damage mechanisms of defects' competition. It is considered that there are two types of defects in real materials: one is the IDs, such as, dislocations, GB and precipitation particles, which are related to the materials themselves; the other is the PDs, including non-metallic inclusions, cavities, segregation and scratches, and so on, which are associated with the technology of production

or processing. As an example of high-strength AISI 4340 steels, the fatigue damage mechanisms with the change of tensile strength are explained well by combining transition of crack initiation sites from surface to inner inclusion, the competition of IDs and PDs and fracture mechanics. For metallic materials, it is concluded that with changing tensile strength, transition of fatigue cracking mode controlled by fatigue mechanism leads to the general fatigue formula.

Acknowledgements

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